

Model Comparison Report

Target: User Activity Classification (Low / Medium / High)

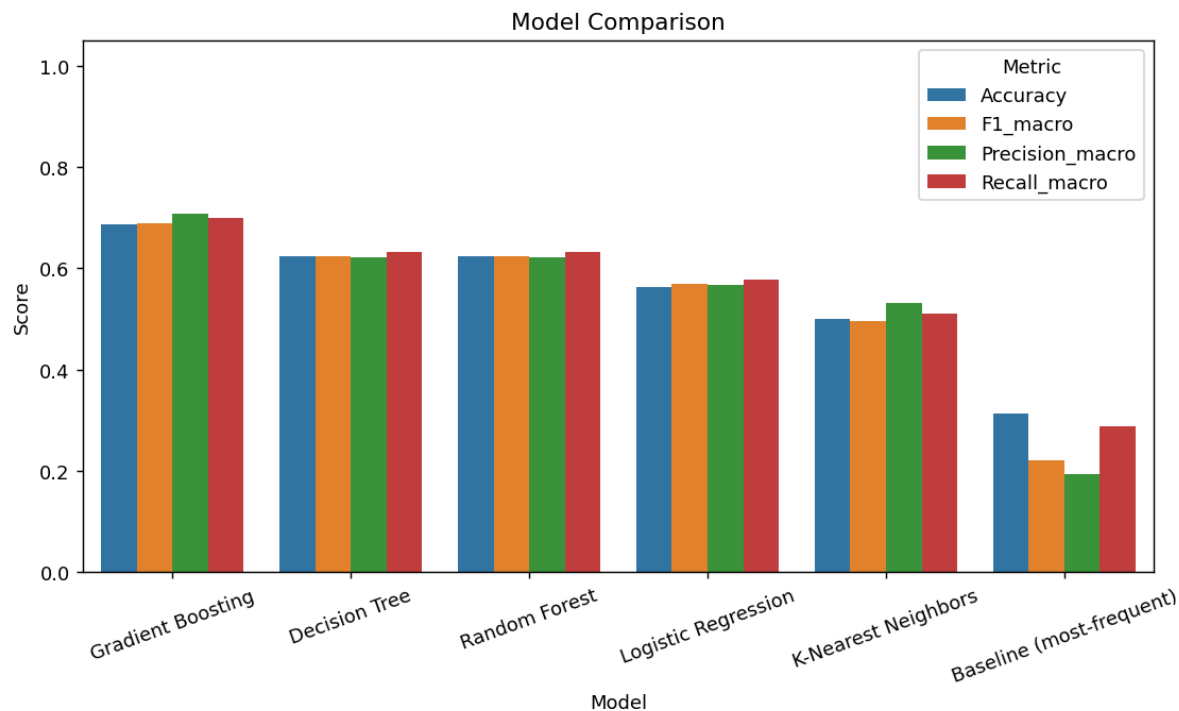
1. Problem Formulation

We frame the task as a 3-class classification: assign each user to a Low, Medium, or High activity tier based on engineered behavioural features. This segmentation supports product personalisation, fraud monitoring focus, and credit-risk pre-screening. Success criterion: macro F1 > baseline (most-frequent classifier).

2. Models Trained

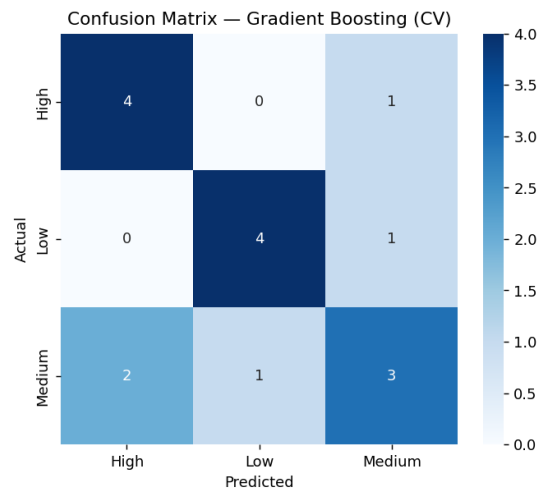
Six classifiers were evaluated using stratified 5-fold cross-validation on 16 users: a most-frequent baseline, Logistic Regression, K-Nearest Neighbors, Decision Tree, Random Forest, and Gradient Boosting.

Model	Accuracy	Precision_macro	Recall_macro	F1_macro
Gradient Boosting	0.688	0.707	0.7	0.689
Decision Tree	0.625	0.622	0.633	0.624
Random Forest	0.625	0.622	0.633	0.624
Logistic Regression	0.562	0.567	0.578	0.57
K-Nearest Neighbors	0.5	0.532	0.511	0.497
Baseline (most-frequent)	0.312	0.194	0.289	0.222



3. Best Model & Tuning

The top performer pre-tuning was **Gradient Boosting**. After Grid-Search CV, the best hyperparameters were {'learning_rate': 0.05, 'max_depth': 2, 'n_estimators': 100} with a macro-F1 of **0.633** (cross-validated). The fitted estimator is exported to *best_model.pkl*.



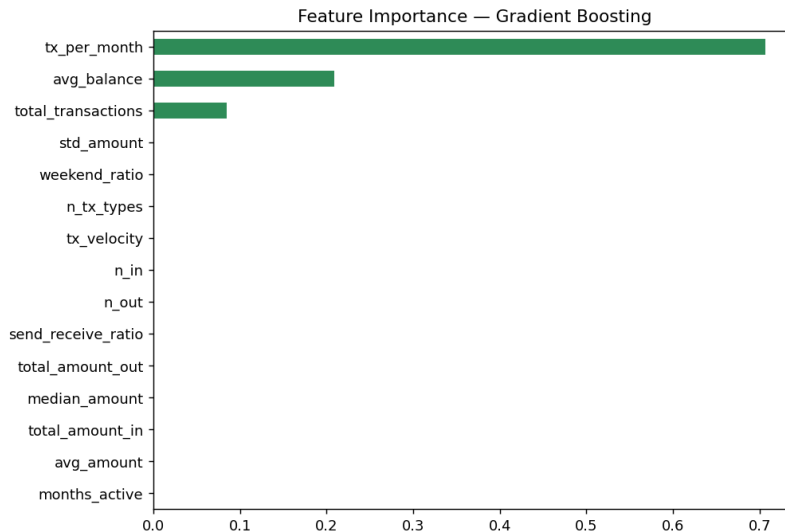
Confusion matrix (cross-validated predictions).

4. Classification Report

Best model: Gradient Boosting

	precision	recall	f1-score	support
High	0.67	0.80	0.73	5
Low	0.80	0.80	0.80	5
Medium	0.60	0.50	0.55	6
accuracy			0.69	16
macro avg	0.69	0.70	0.69	16
weighted avg	0.68	0.69	0.68	16

5. Feature Importance



The most predictive features are **tx_per_month**, **avg_balance**, **total_transactions**, confirming the EDA hypothesis that volume and frequency drive activity classification.

6. Trade-offs

Gradient Boosting offers the best discriminative power but is less interpretable than Logistic Regression or a shallow Decision Tree. Given the small sample ($n=16$), there is real risk of over-fitting; therefore we kept tree depth small (≤ 3) and used cross-validated metrics for all reporting. For deployment with more data, the same architecture should be retrained with calibration.